

Assignment 2

Question 1.

$$v_{z,\text{sun}} = +7 \text{ km/s} , \quad z(t=t_{\text{now}}) = 0 , \quad \text{say } t_{\text{now}} = 0$$

$$\rho(z) = 0.1 M_{\odot} / \text{pc}^3 , \text{ constant} . \quad v_c = \text{constant}$$

- 1a) (9.18) $4\pi G \rho \approx \nu^2$, where ν is the frequency of vertical oscillations. This only applies because the sun's orbit does not stray far from the midplane (told $\rho \sim \text{constant}$)

$$\nu = z \sqrt{\pi G \rho} \xrightarrow{\text{astroph. units}} = 0.75 \cdot \frac{1}{\text{Myr}}$$

$$\text{period } T = \frac{2\pi}{\nu} = \boxed{83.7 \text{ Myr}}$$

- 1b) next maximum will occur $\sim \frac{1}{2}T$ from now ($\sim 40 \text{ Myr}$)

1c) (9.17) $z = z_{\text{max}} \cos(\nu t + \xi)$, $t_{\text{now}} = 0$, $\xi = 0$

$$z(t=0) = 0 = z_{\text{max}} \cos(\xi)$$

$$\text{because } z_{\text{max}} \neq 0, \quad \xi = \pm \frac{1}{2}\pi$$

$$\text{We also know } v_z = \dot{z} = -z_{\text{max}} \sin(\nu t + \xi)$$

$$\text{at } t=0, \quad v_z = 7 \text{ km/s} = -z_{\text{max}} \cdot \nu \sin(\pm \frac{1}{2}\pi)$$

$$\frac{7 \text{ km/s}}{\nu} = z_{\text{max}} \xrightarrow{\text{astroph. units}} \boxed{z_{\text{max}} = 95.21 \text{ pc}}$$

$$\frac{n(z_{\text{max}} = 95.21 \text{ pc})}{n(z_{\text{now}} \approx 0)} = \frac{e^{-95.21 \text{ pc} / 200 \text{ pc}}}{e^0} = 0.62 \quad \text{or } 62\% \quad \text{of the density at } z_{\text{now}}$$

$\therefore$ We would see $\sim 60\%$ less stars in the night sky (but perhaps a brighter MW?)

Question 2

Spherical galaxy, $v_c = \text{constant}$, tracer number density $\nu \propto r^{-1}$

2a) The spherical Jeans equation

$$(5.47) \quad \frac{1}{\nu} \frac{d}{dr} \nu \sigma_r^2(r) + \frac{2\beta \sigma_r^2(r)}{r} = \frac{d\Phi(r)}{dr} = -\frac{GM(r)}{r^2}$$

Since $\beta=0$, the second term disappears.

$$\frac{1}{\nu} \frac{d}{dr} \nu \sigma_r^2(r) = -\frac{GM(r)}{r^2}$$

$$\frac{d}{dr} \nu \sigma_r^2 = -\frac{\nu}{r} v_c^2, \quad \text{using} \quad v_c^2 = \frac{GM(r)}{r}$$

integrate both sides with respect to r ,

$$\nu \sigma_r^2 = v_c^2 \int -\frac{\nu}{r} dr, \quad \text{since } \nu \propto r^{-1}, \text{ so}$$

$$\nu = k r^{-1}$$

$$= v_c^2 \cdot \int -\frac{k}{r^2} dr$$

$$= v_c^2 \cdot -\frac{k}{r} = -v_c^2 \cdot \nu$$

so, $\sigma_r(r) = v_c \rightarrow \text{constant}$

2b) Now we must consider the point-mass potential of a BH

$$\Phi_{\text{BH}} = -\frac{GM_{\text{BH}}}{r},$$

Treat the combined potential as $\Phi_{\text{tot}} = \Phi_{\text{DM}} + \Phi_{\text{BH}}$

Beginning at

$$\begin{aligned}\frac{1}{v} \frac{d}{dr} v \sigma_r^2(r) &= \frac{d\Phi(r)}{dr} = \frac{d}{dr} (\Phi_{\text{DM}} + \Phi_{\text{BH}}) \\ &= \underbrace{\frac{-GM_{\text{DM}}(<r)}{r^2}}_{\frac{v_c^2}{r}} - \frac{GM_{\text{BH}}}{r^2}\end{aligned}$$

integrating both sides

$$\int \frac{d}{dr} v \sigma_r^2(r) dr = - \int v \frac{v_c^2}{r} dr - GM_{\text{BH}} \int \frac{v}{r^2} dr$$

$$v \sigma_r^2(r) = v_c^2 v - GM_{\text{BH}} \int \frac{v}{r^3} dr$$

$$= v_c^2 v + GM_{\text{BH}} \frac{v}{2r^2}$$

$$= v_c^2 v + \frac{GM_{\text{BH}} \cdot v}{2r}$$

$$\rightarrow \sigma_r^2(r) = v_c^2 + \frac{GM_{\text{BH}}}{2r}$$

need to show that $R_{\text{influence}} \sim \frac{GM_{\text{BH}}}{v_c^2}$

BH dominates kinematics when $\frac{GM_{\text{BH}}}{2r} \sim v_c^2$, so,

$$\sigma^2(r_{\text{inf}}) \sim \frac{2GM_{\text{BH}}}{2r_{\text{inf}}} = \frac{GM_{\text{BH}}}{r_{\text{inf}}} = v_c^2 + \frac{GM_{\text{BH}}}{2r_{\text{inf}}}$$

$$\text{then, } \frac{z}{2} \cdot \frac{G M_{BH}}{R_{inf}} - \frac{G M_{BH}}{2 R_{inf}} = v_c^2$$

$$\frac{G M_{BH}}{2 R_{inf}} = v_c^2$$

$$\rightarrow R_{inf} \sim \frac{G M_{BH}}{v_c^2}$$

$$\frac{\sigma^2(r/R_{inf})}{v_c^2} = \frac{v_c^2}{v_c^2} + \frac{G M_{BH}}{2 r / R_{inf}} \frac{1}{v_c^2} \cdot \frac{1}{R_{inf}}$$

$$= 1 + \frac{1}{2 r / R_{inf}}$$

$$\text{so } \frac{\sigma(r/R_{inf})}{v_c} = \sqrt{1 + \frac{1}{2 r / R_{inf}}}$$

plot on next page.

Here, the 'background' is 1, since in the absence of a BH, $\sigma(r) = v_c$.

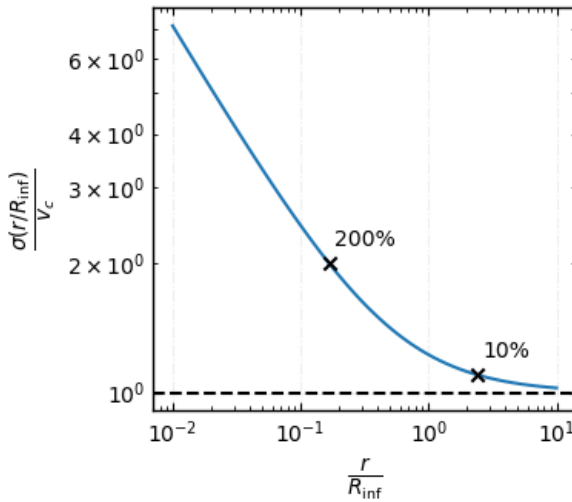
Then, the dispersion profile deviates by 10% and a factor of 2 when

$$\frac{\sigma(r/R_{inf})}{v_c} = 0.1 \text{ and } 2 \text{ respectively.}$$

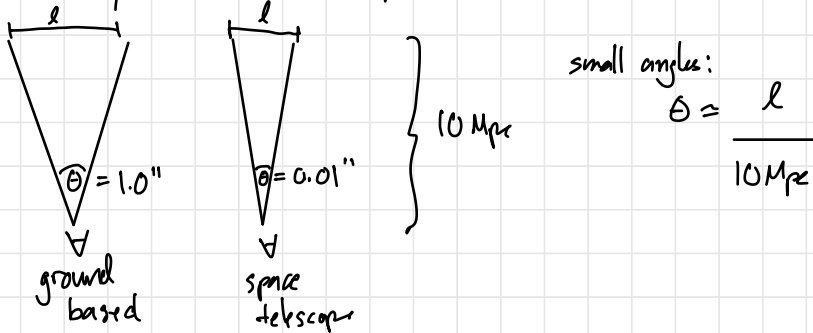
For some increase factor X ,

$$X^2 = 1 + \frac{1}{2 r / R_{inf}} \rightarrow ((X^2 - 1)^2 \cdot 4)^{-1} = r / R_{inf}$$

so we find 10% increase at $r/R_{\text{inf}} = 2.3$
 and a 200% increase at $r/R_{\text{inf}} = 1/6$.



2c) Galaxy distance 10 Mpc, $v_c = 200 \text{ km s}^{-1}$



$$\theta_g = 1.0'' = 4.9 \mu\text{rad} \rightarrow l_g = 49 \text{ pc}$$

$$\theta_s = 0.01'' = 0.049 \mu\text{rad} \rightarrow l_s = 0.49 \text{ pc}$$

The space telescope can resolve physical scales $\sim 0.5 \text{ pc}$, while the ground based telescope resolves $\sim 50 \text{ pc}$ scales.

For a black hole to be detectable, we would need to be able to resolve the deviation of σ_r^2 from the no-BH scenario.

That is, we can connect R_{inf} to the spatial resolutions of these telescopes.

For HST, say $l = 0.5 \text{ pc} \sim R_{\text{inf}}$

$$\text{then, } M_{\text{BH}} = \frac{v_c^2 \cdot R_{\text{inf}}}{G} \sim \text{astroph. units } \underline{4 \times 10^6 M_\odot}$$

$\uparrow$
200 km/s
given

For ground based, $M_{\text{BH}} \sim \underline{4 \times 10^8 M_\odot}$

Respectively, these are the minimum detectable masses for the two telescopes.

2d)

We would require an instrument capable of measuring the velocity dispersions within nearby galaxies' galactic centers. To this end, we would want to spatially resolve the kinematics of stars within the sphere of influence set by R_{inf} .

Photometry alone isn't sufficient; to estimate the line-of-sight velocities of the spatially resolved stars, we must also have the spectral resolution to measure the doppler shifts of key spectral features. The amount of red/blue shifting occurring is tied to the motion of the star with respect to us, the observer.

In broad strokes, ground based facilities (e.g. VLT, ELT) are capable of achieving higher spectral resolutions with their spectrographs (MUSE/HARMONI). The limiting factor with regard to ground based facilities is that observations from the ground must see through the atmosphere. Turbulence in the atmosphere affects "seeing", causing a blurring out of the smallest scales one can resolve.

The atmosphere is not an issue for space based telescopes (JWST, HST); the limitations are rather with respect to the cargo space/engineering difficulties of launching the telescope. This restricts the size/complexity of on-device instruments (like spectrographs).

Question 3.

Rotation Curve Fitting using SPARC sample of 175 nearby galaxies.

- photometry in mid-infrared
- 21cm and H α observations

Code has function `plot-rotcurve-model` that takes arguments

`RRange`: radii

`disk-surfden`: central surface density of disk in $M_{\odot}/pc^2$

`disk-hr`: disk scale length

`halo-virtual-mass`: NFW virtual mass in $M_{\odot}$

`halo-conc`: NFW concentration parameter.

Notebook attached.